DATABASES

Laboratory work 5

Constraints and default values

Student

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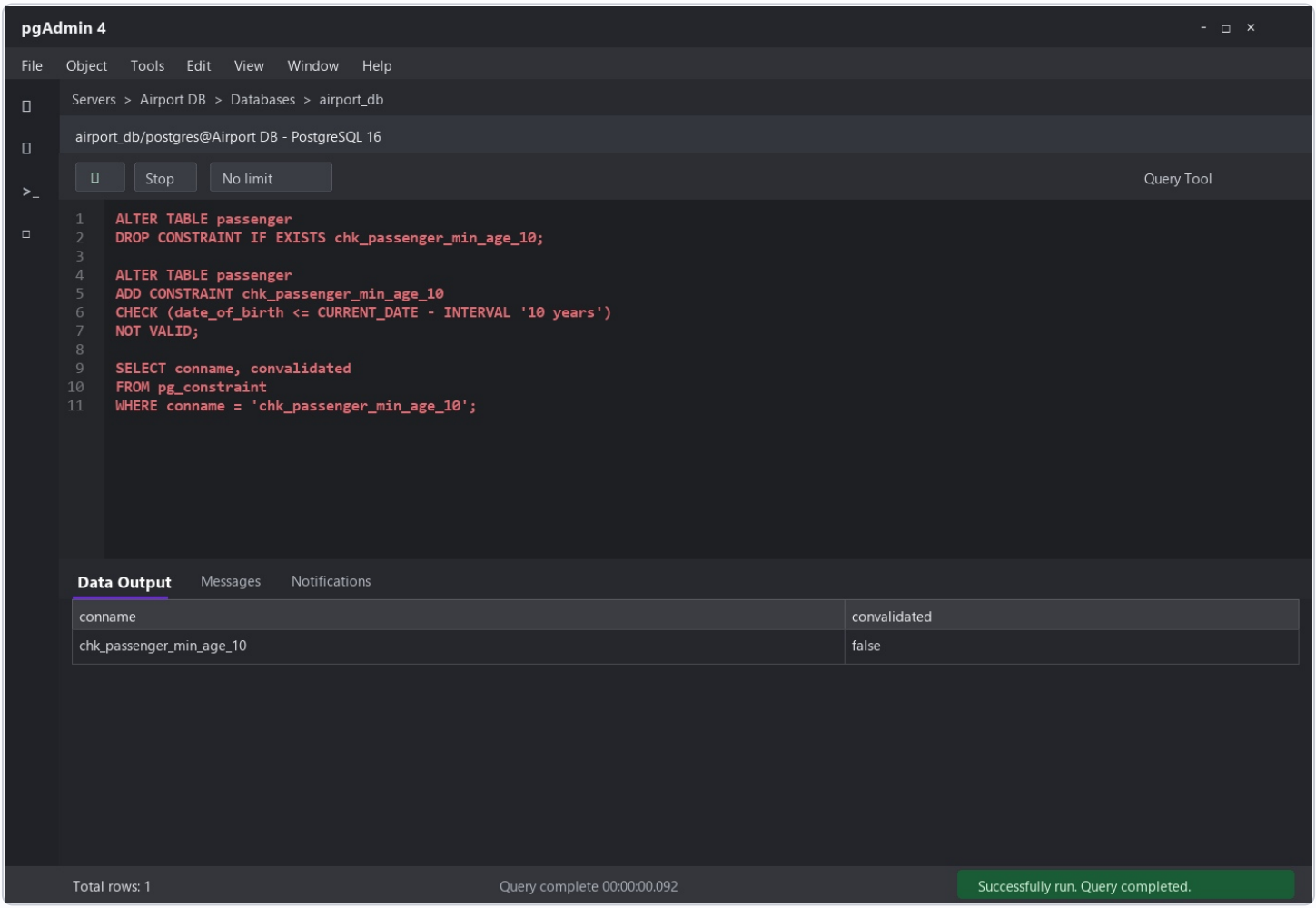
Database airport_db

PostgreSQL 16

05

Task 1

Passengers must be at least 10 years old.

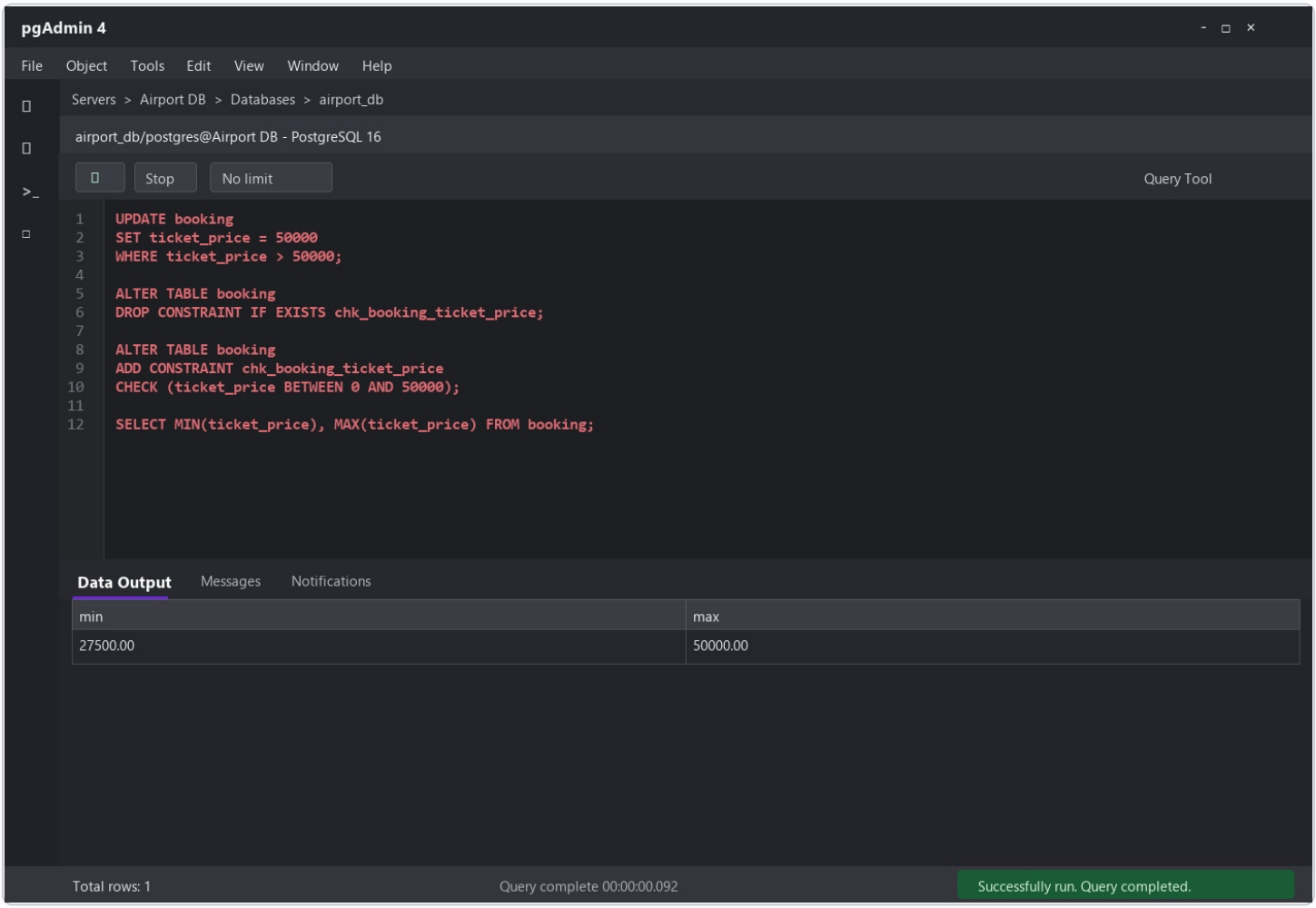


COMMAND EXPLANATION

The CHECK compares date_of_birth with the date exactly ten years ago. NOT VALID keeps old rows untouched, but PostgreSQL still applies the rule to every new or updated passenger.

Task 2

Accept booking prices from 0 to 50000 tenge.

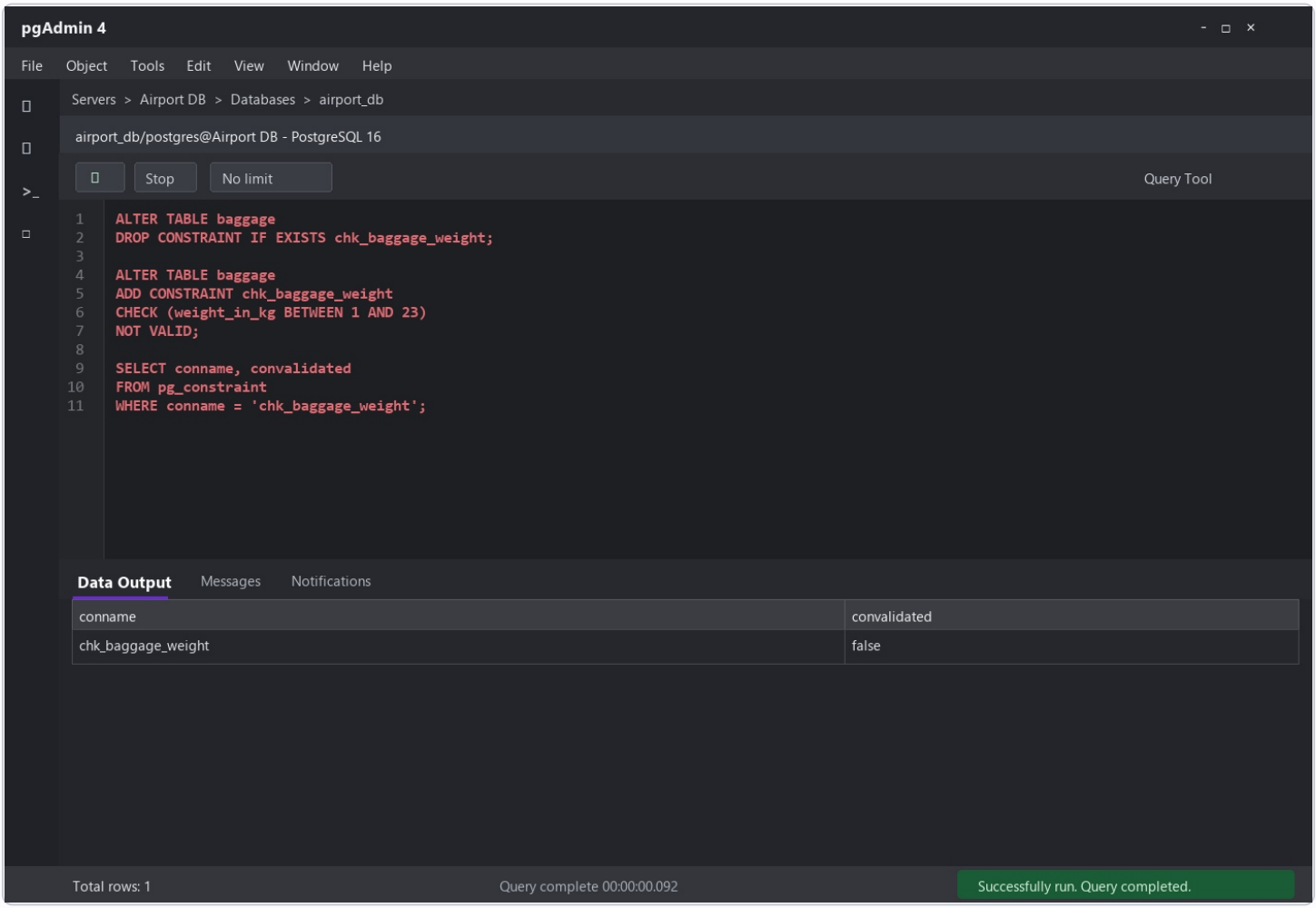


COMMAND EXPLANATION

Existing prices above the limit are corrected before the validated CHECK is added. BETWEEN includes both 0 and 50000. The phrase less than 0 in the assignment is interpreted as not less than 0.

Task 3

Accept baggage weight from 1 to 23 kg.

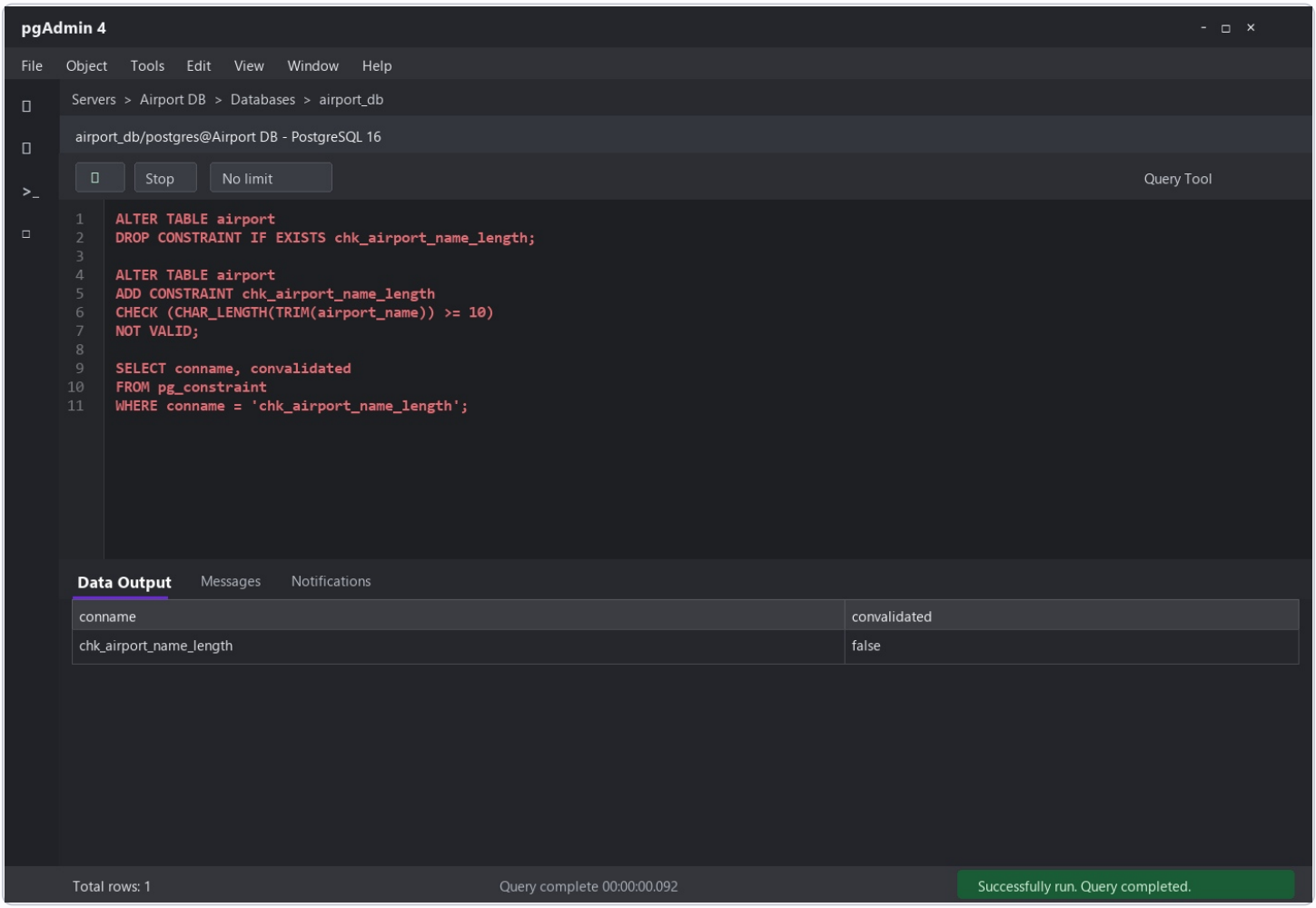


COMMAND EXPLANATION

BETWEEN makes the allowed interval inclusive. NOT VALID preserves baggage rows from the previous lab that are heavier than 23 kg, while invalid new or changed rows are rejected.

Task 4

Require at least 10 characters in airport_name.

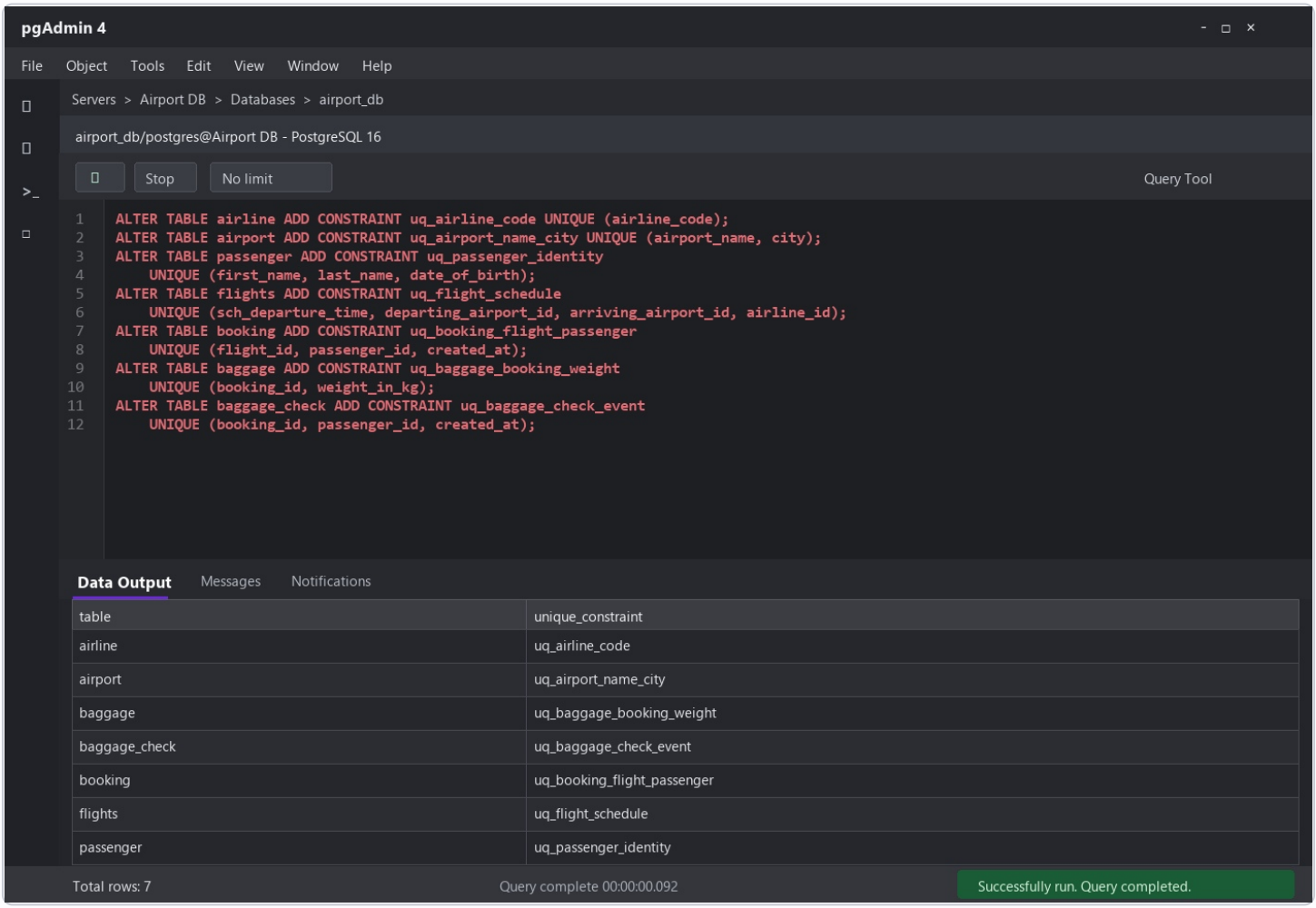


COMMAND EXPLANATION

TRIM removes spaces at the beginning and end, and CHAR_LENGTH counts the remaining characters. The CHECK accepts an airport name only when the length is at least ten.

Task 5

Add UNIQUE constraints to the database tables.

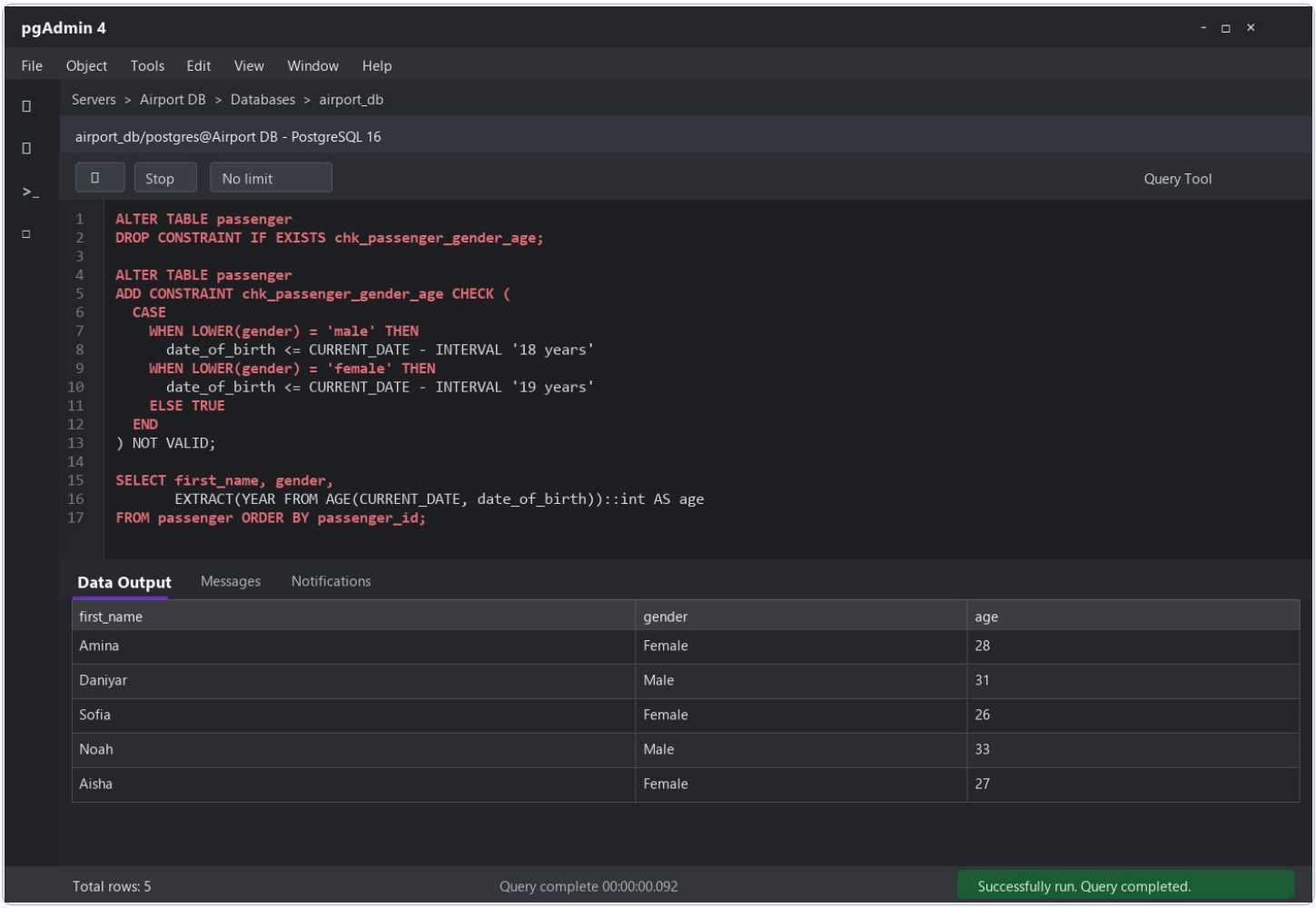


COMMAND EXPLANATION

Each UNIQUE rule protects a natural combination of columns, not only the primary key. PostgreSQL creates a unique index for every rule and rejects future duplicate values.

Task 6

Apply different minimum ages for male and female passengers.

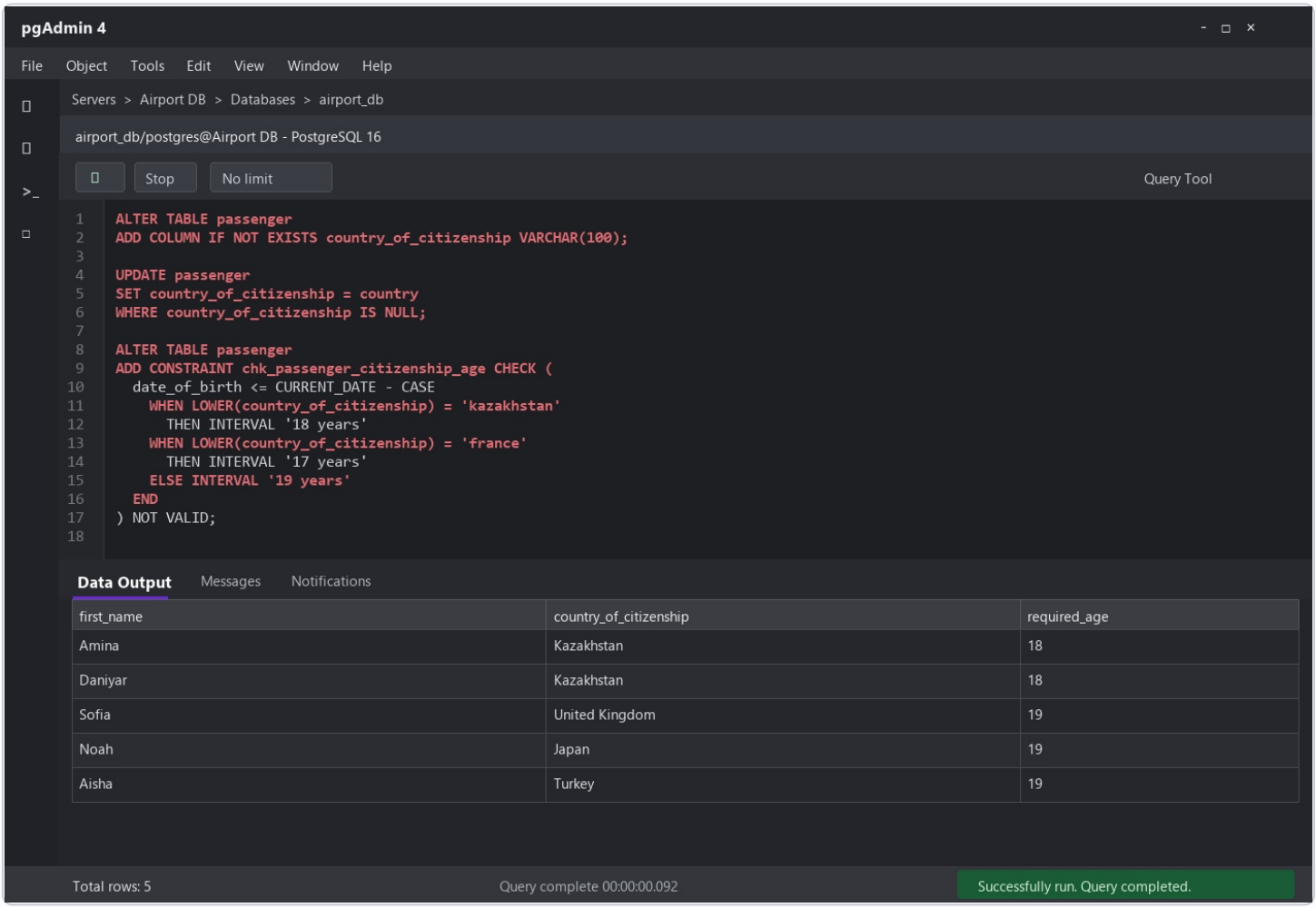


COMMAND EXPLANATION

CASE chooses the correct age limit from gender. LOWER makes the comparison case-insensitive. Other gender values are not restricted by this particular rule.

Task 7

Apply age limits based on country of citizenship.

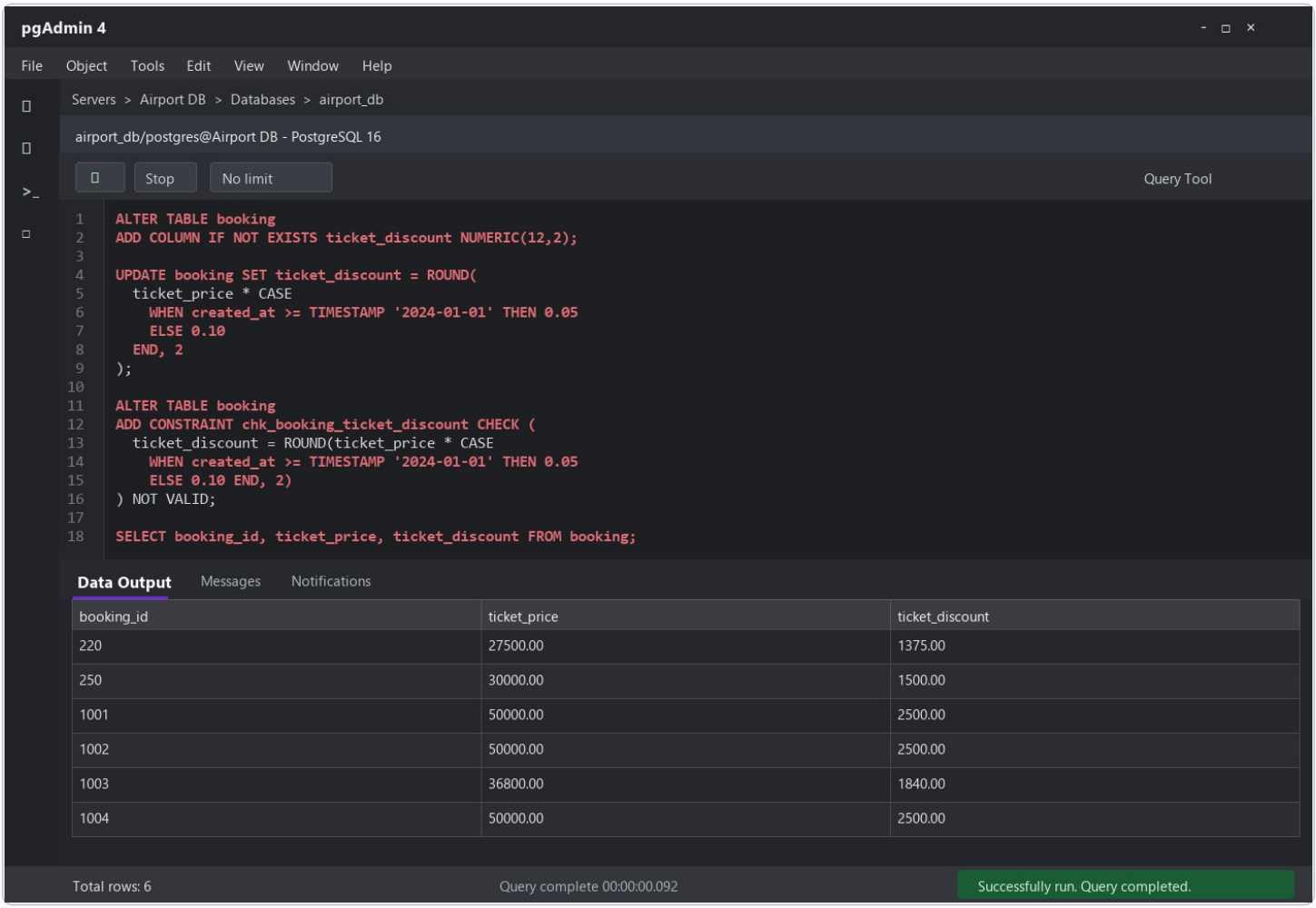


COMMAND EXPLANATION

The current database used country, so country_of_citizenship is added and filled from it. CASE sets 18 for Kazakhstan, 17 for France and 19 for every other country.

Task 8

Add ticket_discount and validate it from created_at.

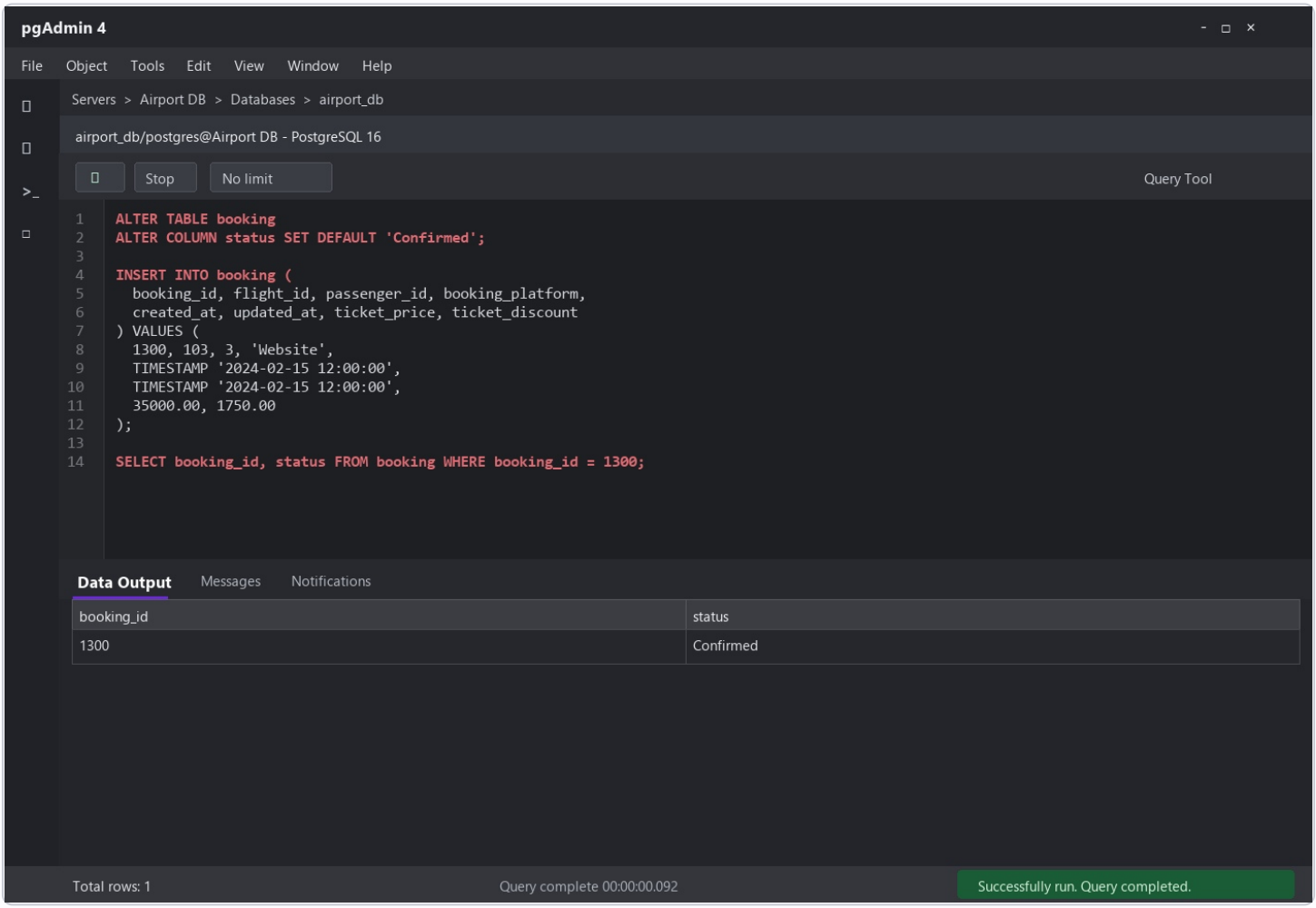


COMMAND EXPLANATION

The discount column stores the amount, not the final price. Rows created on or after 2024-01-01 receive 5 percent; older rows receive 10 percent. ROUND keeps two decimal places.

Task 9

Set Confirmed as the default booking status.

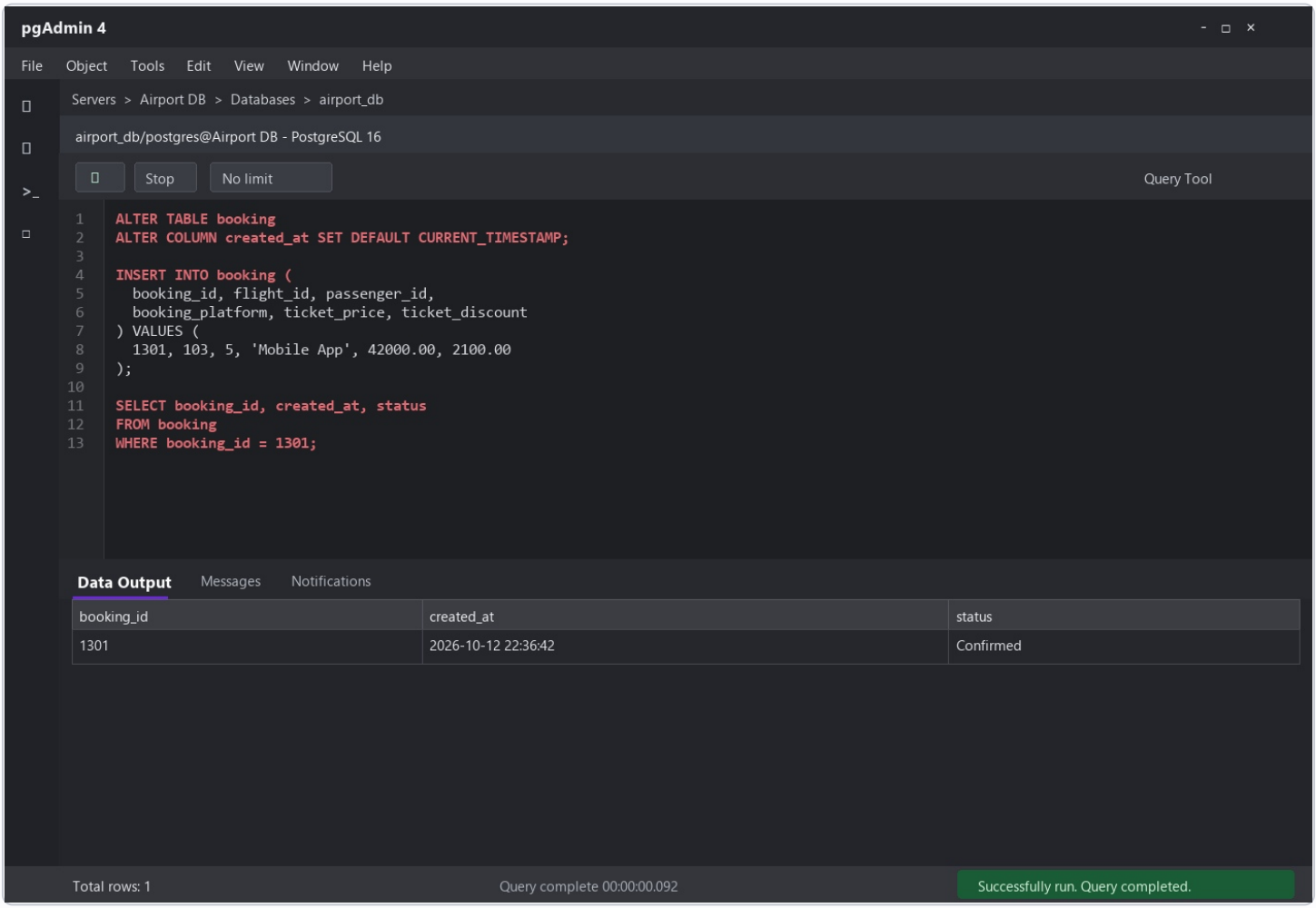


COMMAND EXPLANATION

SET DEFAULT supplies Confirmed whenever status is omitted. The INSERT does not include the status column, and the result verifies that PostgreSQL added the value automatically.

Task 10

Set the current time automatically for new bookings.



COMMAND EXPLANATION

CURRENT_TIMESTAMP becomes the default for created_at. The INSERT omits both created_at and status; PostgreSQL fills the current timestamp and the default Confirmed status.